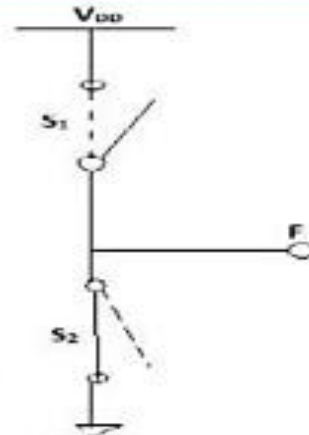


CMOS UNIT III & IV

General Logic Circuit



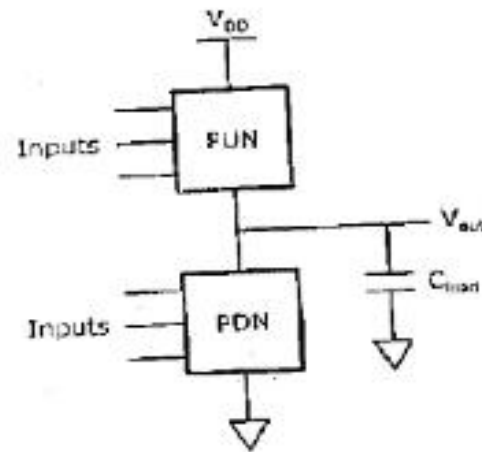
- The switch S_1 is closed and switch S_2 is open for input combinations that produces $F = 1$.
- The switch S_1 is open and switch S_2 is closed for input combinations that produces $F = 1$.
- The switch S_1 is open and switch S_2 is open for input combinations that produces $F = 0$.

Thus the output (F) is either connected to V_{DD} or the ground, where the logic 0 is represented by the ground and the logic 1 is represented by V_{DD} . So the requirement of digital logic design is to implement the pull-up switch(S_1) and the pull-down switch(S_2).

CMOS STATIC LOGIC



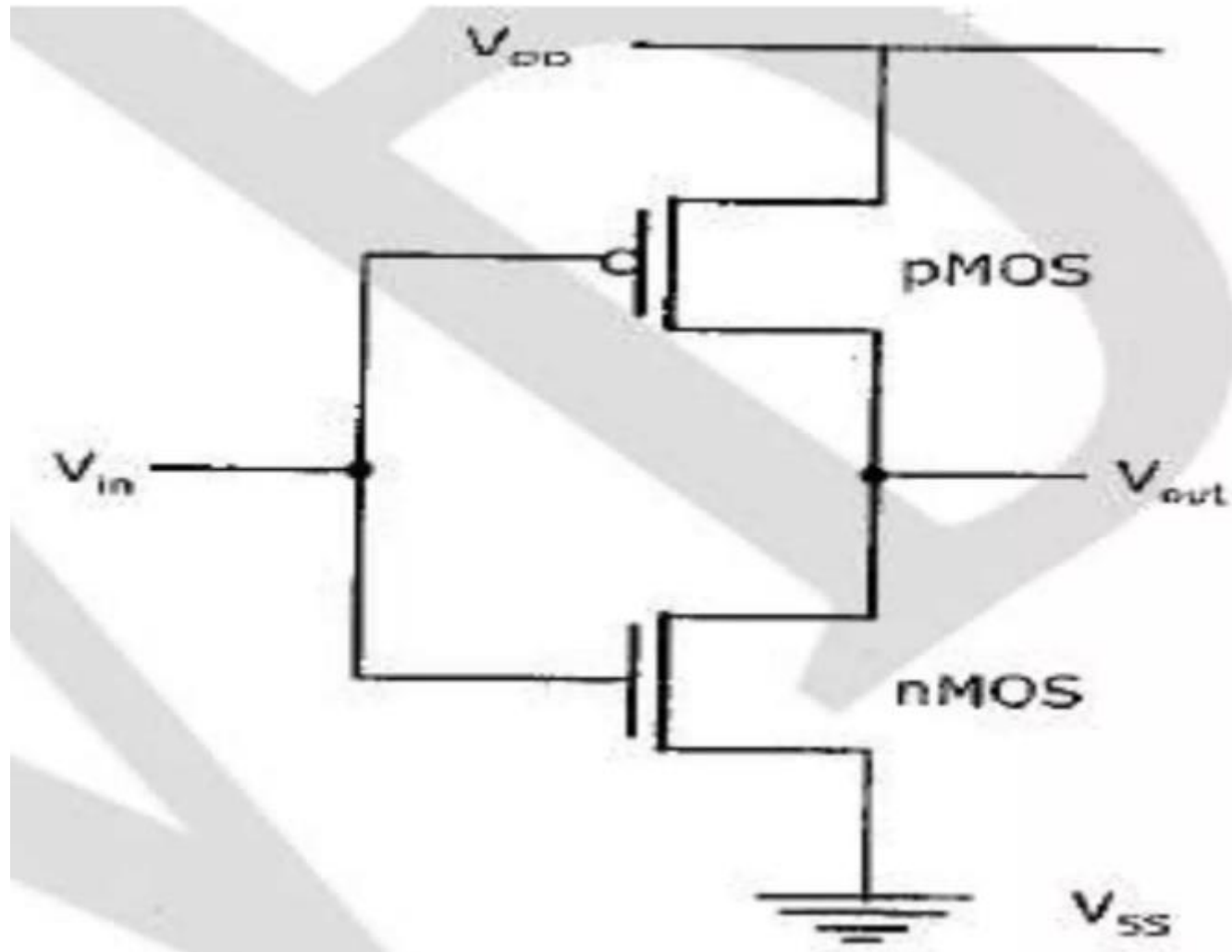
A generalized CMOS logic circuit consists of two transistor nets nMOS and pMOS. The pMOS transistor net is connected between the power supply and the logic gate output called as pull-up network , Whereas the nMOS transistor net is connected between the output and ground called as pull-down network. Depending on the applied input logic, the PUN connects the output node to V_{DD} and PDN connects the output node to the ground.



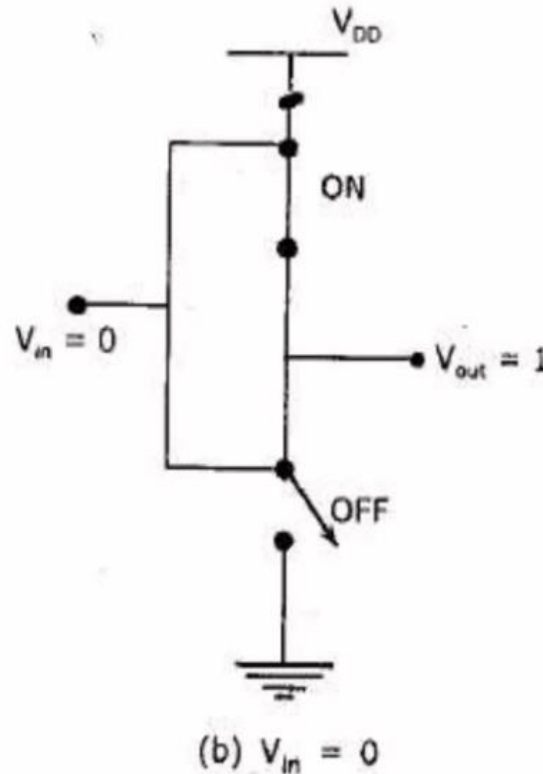
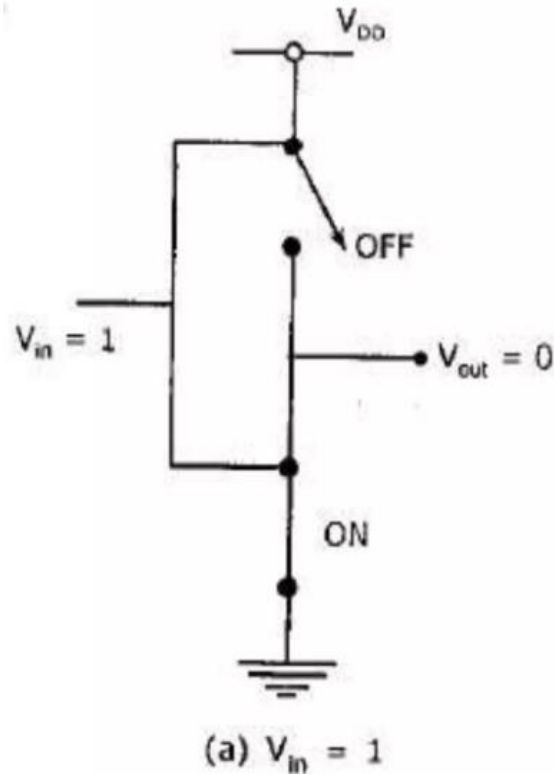
The transistor network is related to the Boolean function with a straight forward design procedure:

- Design the pull down network (PDN) by realizing, AND(product) terms using series-connected nMOSFETs. OR (sum) terms using parallel-connected nMOSFETs.
- Design the pull-up network by realizing, AND(product) terms using parallel-connected nMOSFETs. OR (sum) terms using series-connected nMOSFETs.
- Add an inverter to the output to complement the function. Some functions are inherently negated, such as NAND, NOR gates do not need an inverter at the output terminal.

CMOS Inverter



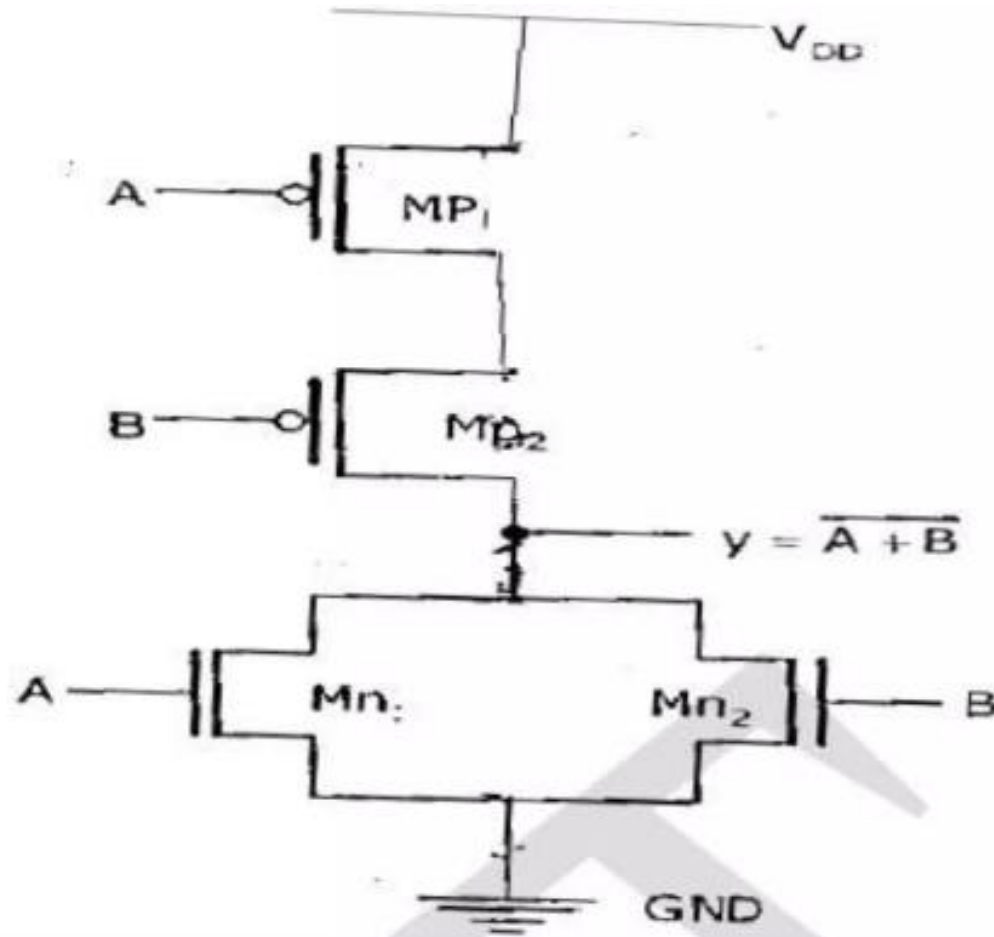
WORKING OF CMOS Inverter



Explanation of Working Operation

- When the input V_{in} is logic HIGH, then the nMOS transistor is ON and the pMOS transistor is OFF. Thus the output Y is pulled down to ground (logic 0) since it is connected to ground but not to source V_{DD} .
- When the input V_{in} is logic LOW, then nMOS transistor is OFF and the pMOS transistor is ON. Thus the output Y is pulled up to V_{DD} (logic 1) since it is connected to source via pMOS but not to ground.

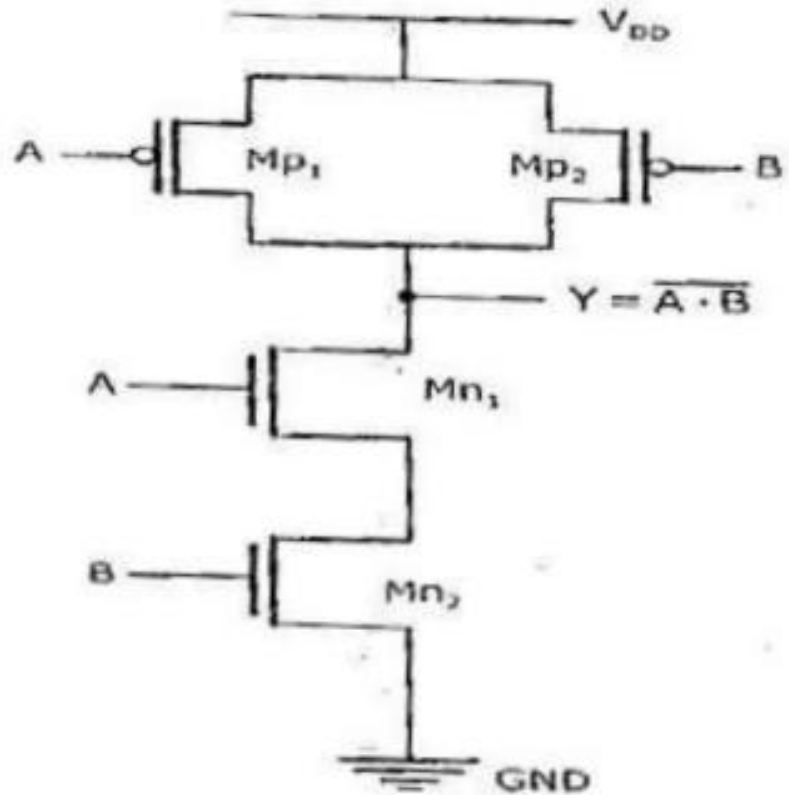
CMOS NOR GATE



Working of CMOS NOR Gate

- Whenever at least one of the inputs is LOW, the corresponding pMOS transistor will conduct while the corresponding nMOS transistor will turn OFF. Subsequently, the output voltage will be HIGH.
- Conversely, if both inputs are simultaneously HIGH, then both pMOS transistors will turn OFF, and the output voltage will be pulled LOW by the two conducting nMOS.

CMOS NAND GATE



Working of CMOS NAND Gate

- Whenever at least one of the inputs is LOW, the corresponding pMOS transistor will conduct while the corresponding nMOS transistor will turn OFF. Subsequently, the output voltage will be HIGH.
- Conversely, if both inputs are simultaneously HIGH, then both pMOS transistors will turn OFF, and the output voltage will be pulled LOW by the two conducting nMOS transistors.

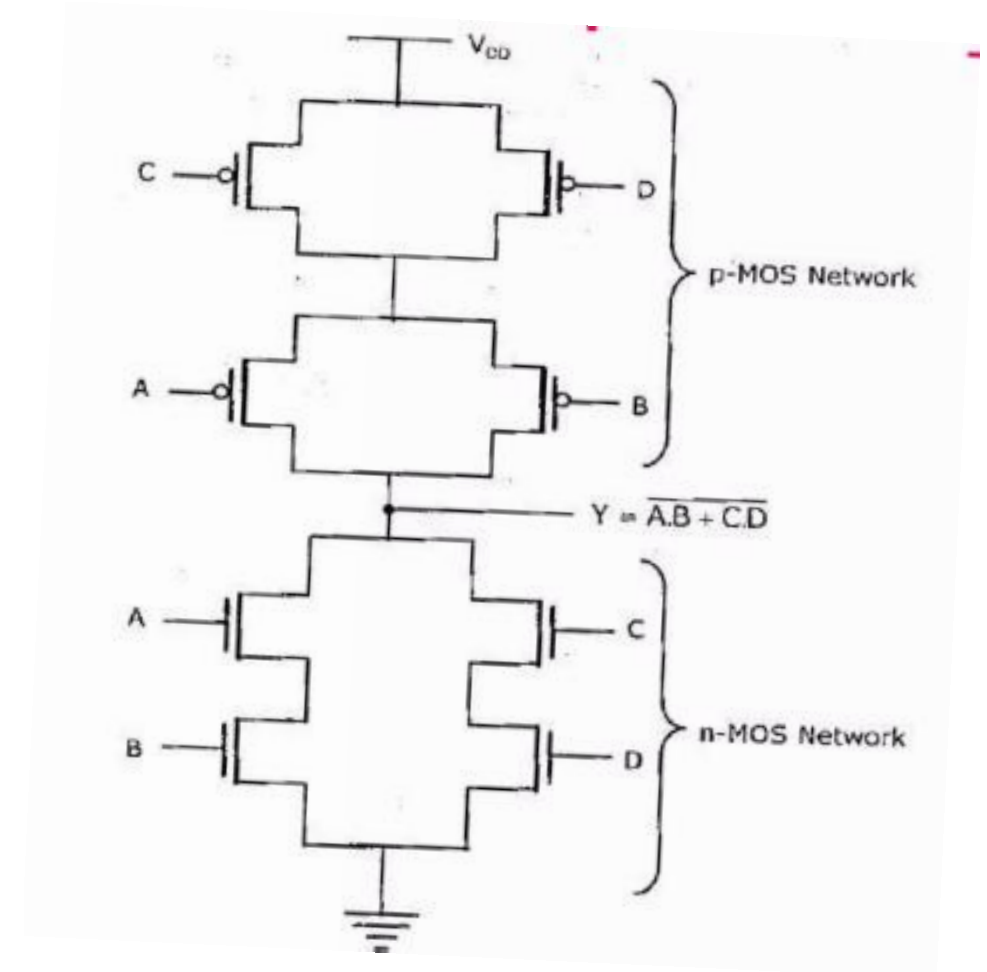
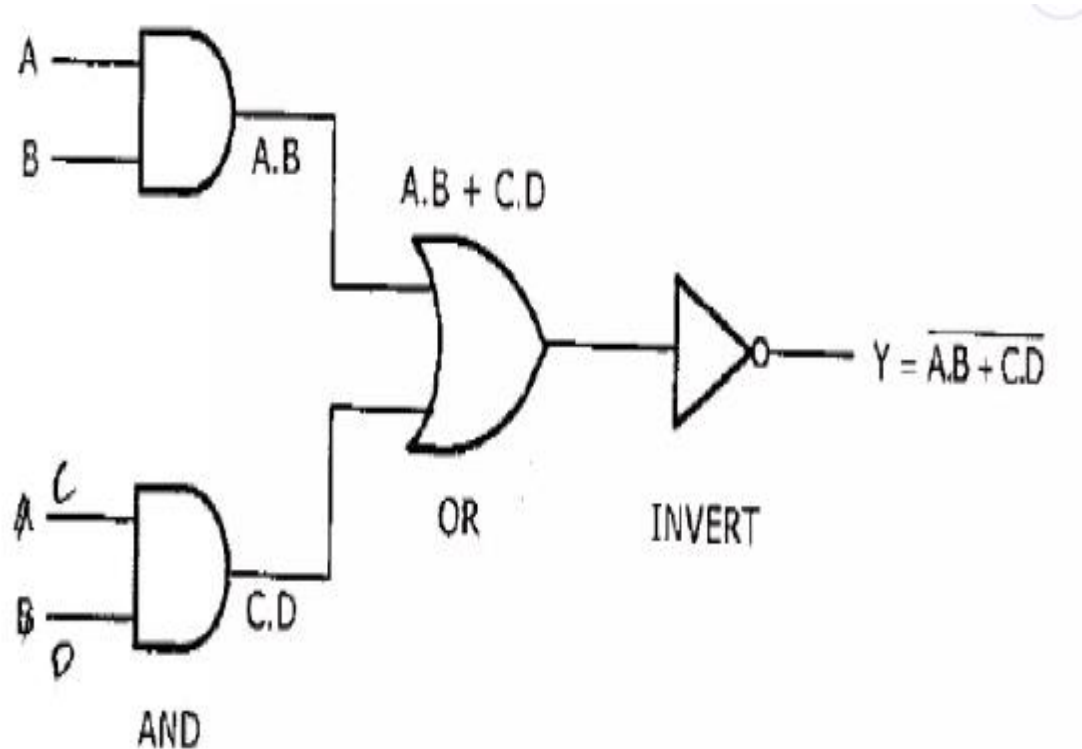
Complex GATE in CMOS logic

A complex logic gate is one that implements a function that can provide the basic NOT, AND and OR operation but integrates them into a single circuit. CMOS is ideally suited for creating gates that have logic equations by exhibiting the following,

- 1.AND-OR-INVERT - AOI form
- 2.OR-AND-INVERT - OAI form

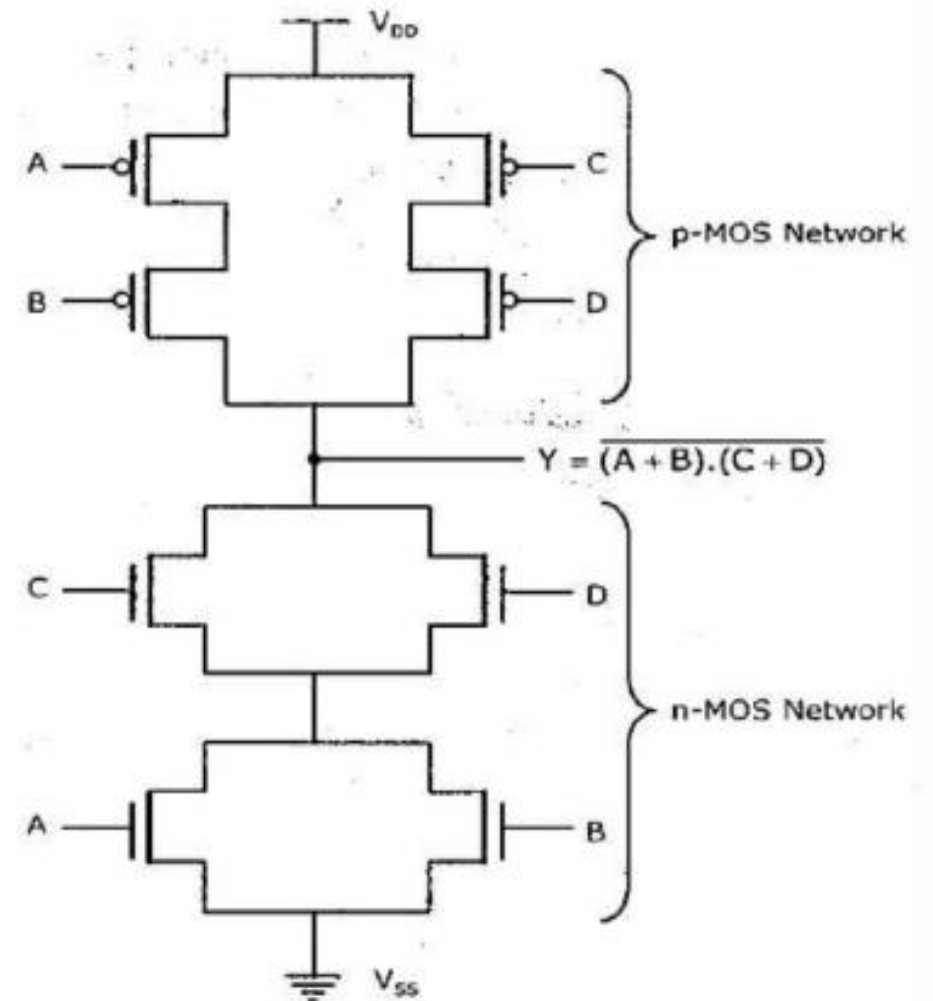
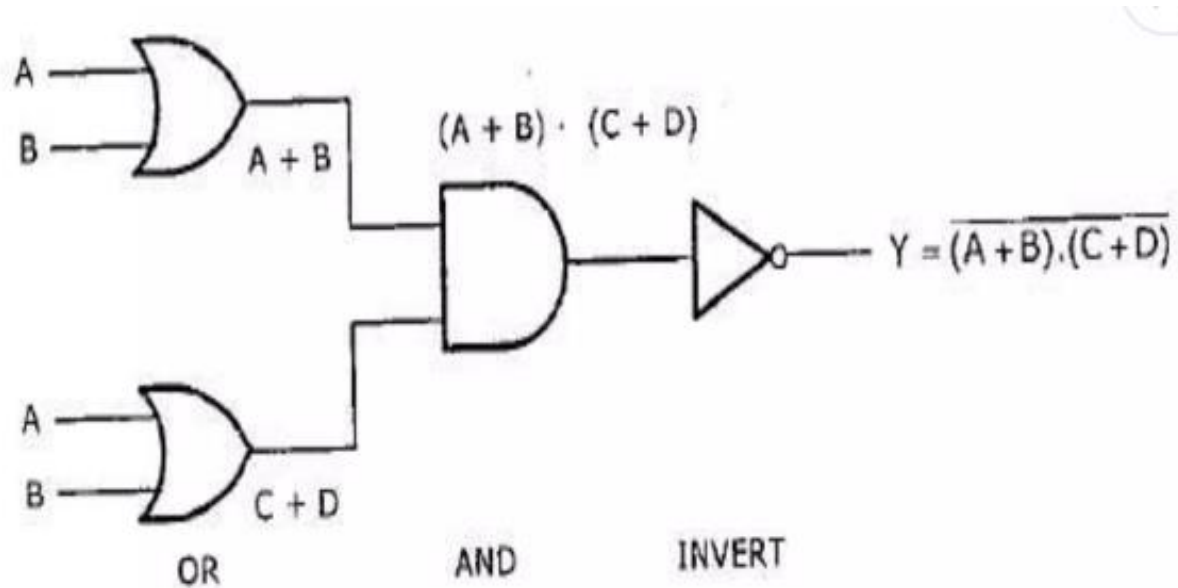
AOI Logic Function or design of XOR GATE using CMOS logic

CMOS Implementation



OAI Logic Function or design of XNOR GATE using CMOS logic

CMOS Implementation

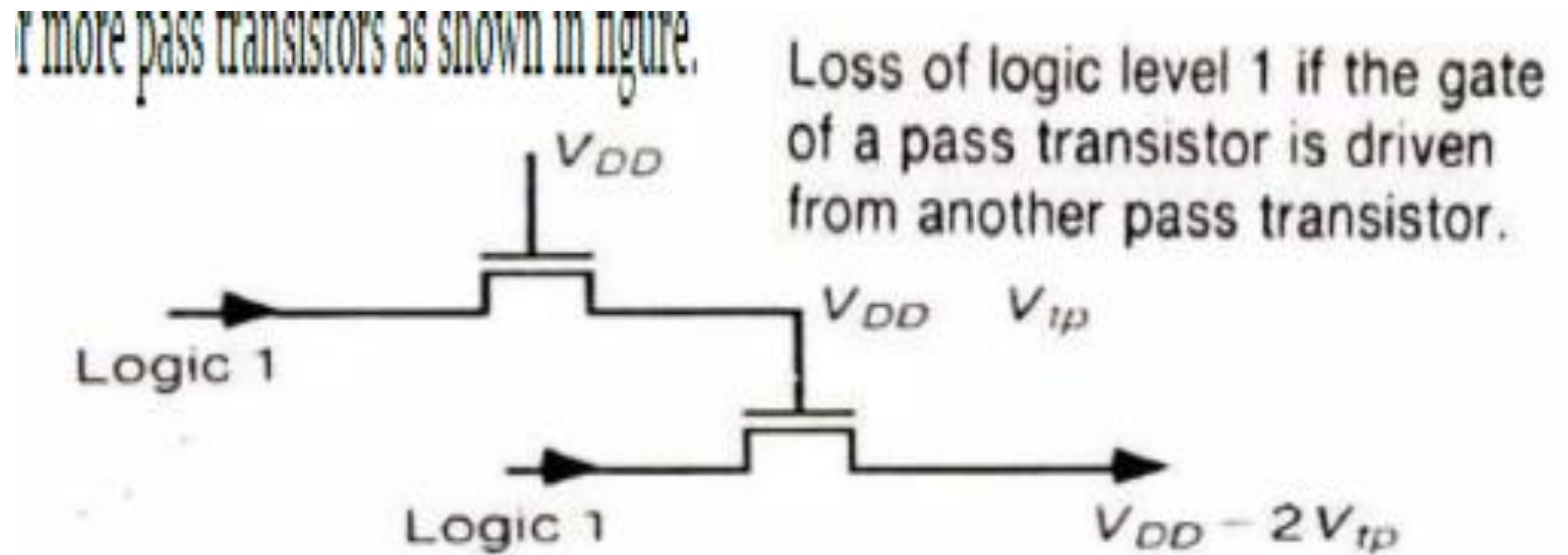


Switch Logic

1. Switch logic is mainly based on pass transistor or transmission gate.
2. It is fast for small arrays and takes no static current from the supply, V_{DD} . Hence power dissipation of such arrays is small since current only flows on switching.
3. Switch (pass transistor) logic is analogous to logic arrays based on relay contacts, where in path through each switch is isolated from the logic levels activating the switch.

Pass Transistor

1. This logic uses transistors as switches to carry logic signals from node to node instead of connecting output nodes directly to V_{DD} or ground(GND)
2. If a single transistor is a switch between two nodes, then voltage degradation is equal to V_{th} (threshold voltage) for high or low level depends up on nMOS or pMOS logic.
3. When using nMOS switch logic no pass transistor gate input may be driven through one or more pass transistors as shown in figure.



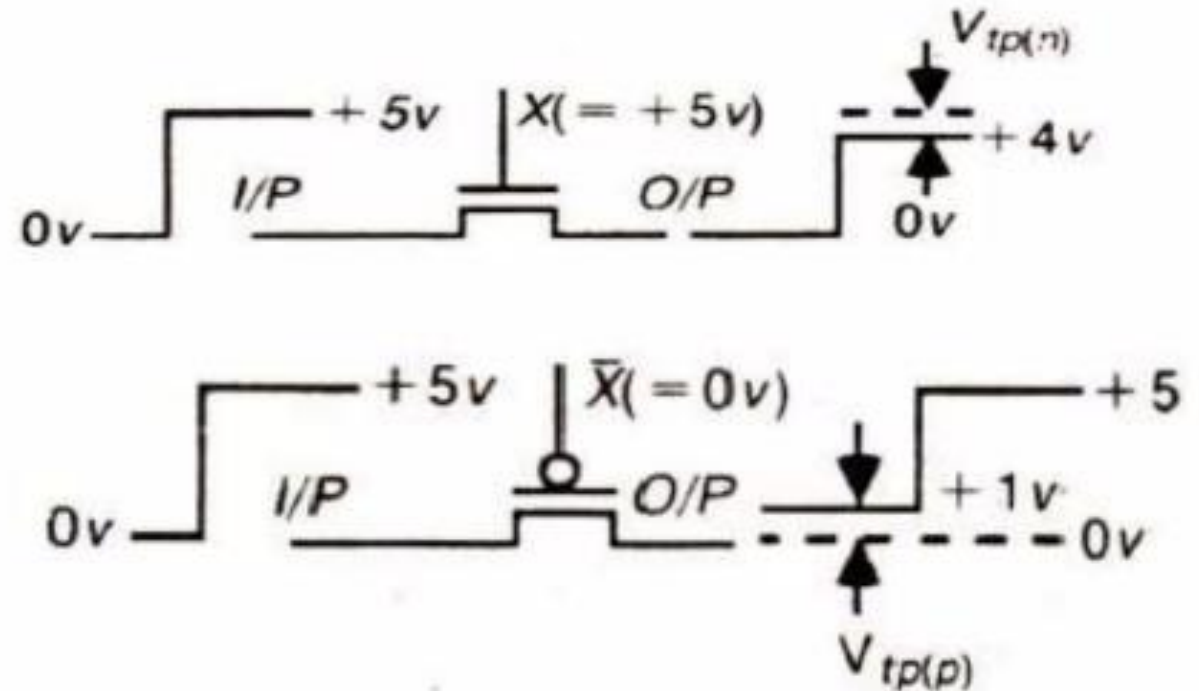
4. Since the signal out of pass transistor T_1 does not reach a full logic 1 by threshold voltage effects, signal is degraded by below a true logic 1, this degraded voltage would not permit the output of T_2 to reach an acceptable logic 1 level.

(Diagram 1: nMOS Transistor)

- Input (I/P) steps from 0V to +5V.
- Gate is held at +5V.
- Output (O/P) is labeled $X(= +5V)$, but the signal trace only reaches +4V. A gap labeled $V_{tp(n)}$ is shown between +4V and +5V.

(Diagram 2: pMOS Transistor)

- Input (I/P) steps from +5V to 0V.
- Gate is held at 0V.
- Output (O/P) is labeled $\bar{X}(= 0V)$, but the signal trace only drops to +1V. A gap labeled $V_{tp(p)}$ is shown between +1V and 0V.

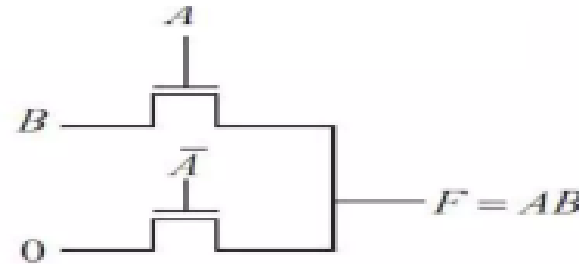


Pass Transistor Logic(PTL)

- Only N_MOS Transistors are Used to design the logic.
- Input Signals are Applied to both Gate and Drain/Source.

A	B	$F = AB$	F
0	0	0	When $A = 0$, $F = 0$
0	1	0	
1	0	0	When $A = 1$, $F = B$
1	1	1	

(a)



(b)

Fig. 6.54 (a) Truth table of two-input AND gate; (b) realization of two-input AND gate using pass transistor logic

When $A=1$ $\longrightarrow$ Upper NMOS is ON, So
o/p is $F=B$

When $A=0$ $\longrightarrow$ Upper
NMOS is ON, So o/p is $F=B$

Pass Transistor Advantages

- 1.Requires minimum geometry.
- 2.Do not dissipate standby power, since they do not have a path from supply to ground.

Disadvantages

- 1.Degradation in the voltage levels due to undesirable threshold voltage effects.
- 2.Never drive a pass transistor with the output of another pass transistor.

CMOS Transmission Gate

- Actually, It is a Parallel Connection of N_MOSFET and P-MOSFET that realizes a simple switch.
- Inputs to the gates of N-MOSFET and P-MOSFET are Complementary to each other.

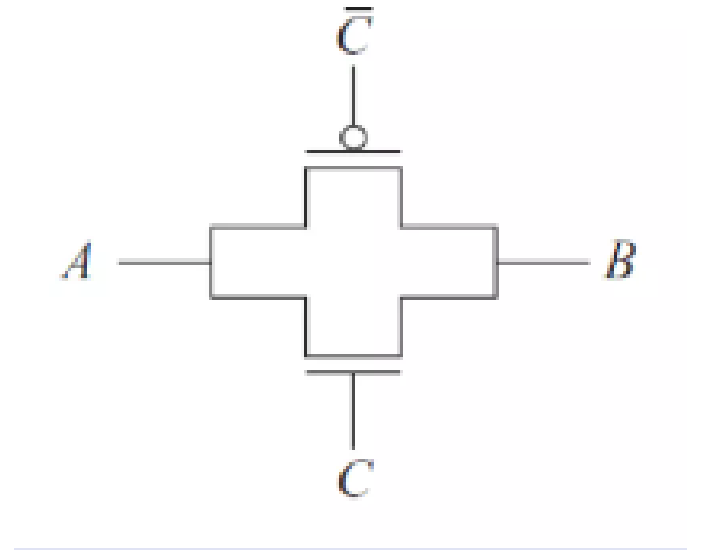
Working of CMOS Transmission Gate

Case-1 When $C=1$ and $\bar{C}=0$

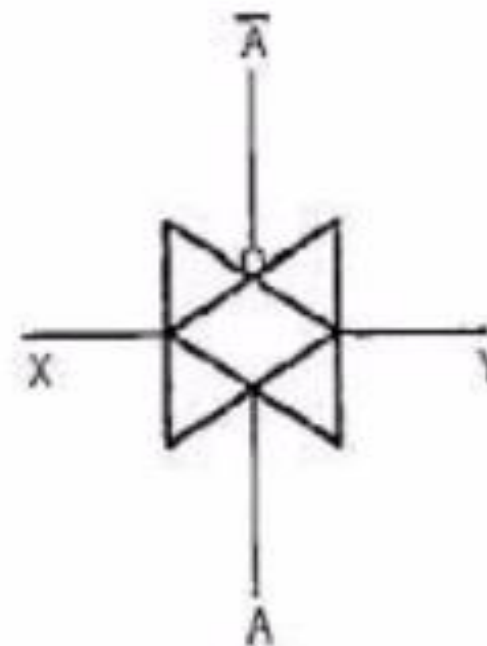
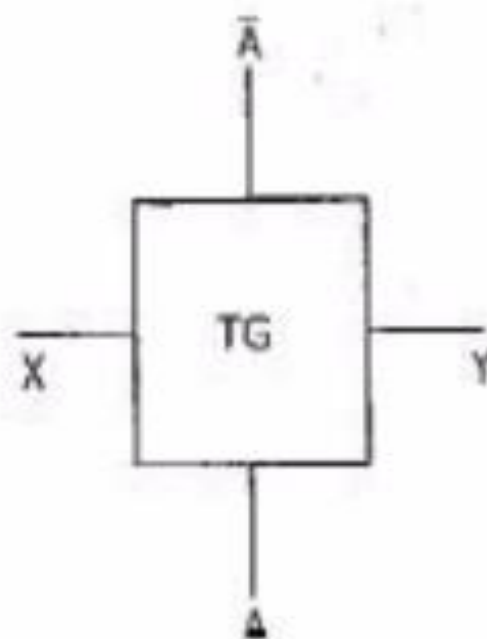
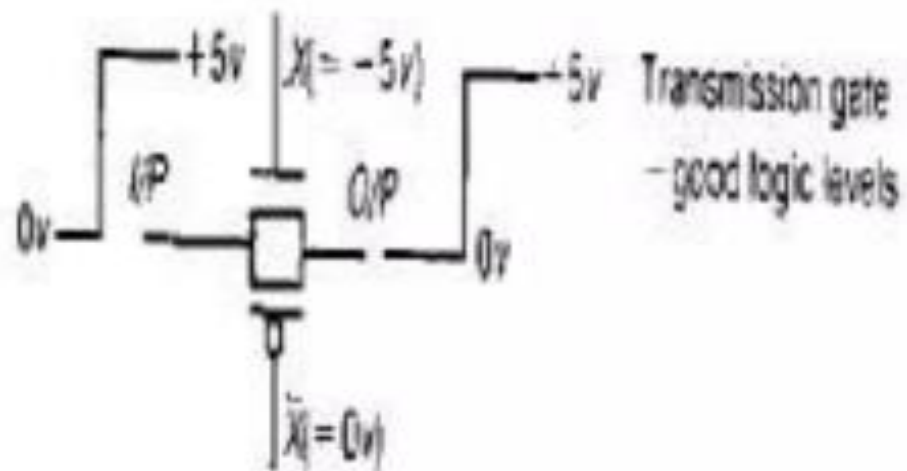
- Both NMOS&PMOS-----ON
- Nodes A&B-----Short Circuited
- Input Logic is Transferred to Output

Case-2 When $C=0$ and $\bar{C}=1$ (Note: This is a typo and should be $C=0$)

- Both NMOS&PMOS-----OFF
- Nodes A&B-----Open Circuited
- Actually ,this is Called High-Impedance State.



Transmission Gate Symbol



Operation of Transmission Gate

- When the gate input to the nMOS transistor is '0' and the complementary '1' is gate input to the pMOS, thus both are turned off.
- When gate input to the nMOS is '1' and its complementary '0' is the gate input to the pMOS, both are turned on and passes any signal '1' and '0' equally without any degradation.
- The use of transmission gates eliminates the undesirable threshold voltage effects which give rise to the loss of logic levels in pass-transistors as shown in figure.

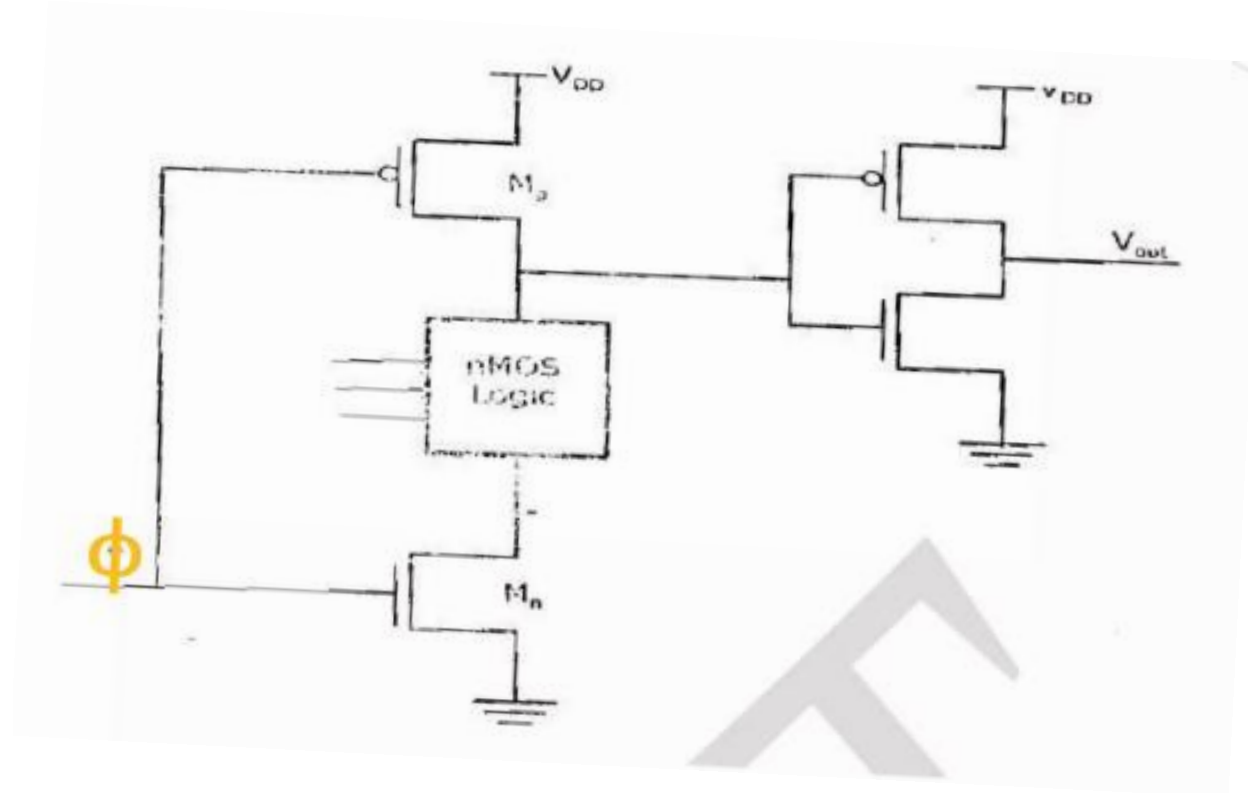
Advantages

1. Transmission gates eliminate the signal degradation in the output logic levels.
2. Transmission gate consists of two transistors in parallel and except near the positive and negative rails.

Disadvantages

1. Transmission gate requires more area than nMOS pass circuitry.
2. Transmission gate requires complemented control signals.

CMOS Domino Logic(Dynamic + Inverter)



Description

- It is a Slightly modified Version of Dynamic CMOS Logic Circuit.
- A Static Inverter is Connected at the Output of each dynamic CMOS logic blocks.
- Addition of Inverter Solves the Problem of Cascading of dynamic CMOS logic Circuits.
- It is Suitable for only Non-Inverting Logic(the expression having no complement over whole expression)
- For Inverting the logic the expression must be reorganized before it can be realized using Domino CMOS Logic.

•Phase 1: Pre-charge ($\phi = 0$)

- When the clock is low, the dynamic part of the circuit turns ON its top PMOS transistor, charging the internal node to a logic HIGH (1).
- Because this internal node is connected to the input of the static inverter, the final output of the entire Domino gate is forced **LOW (0)**.
- Think of this as setting up the dominos before knocking them down. Every gate in the chain resets to a 0 output.

•Phase 2: Evaluation ($\phi = 1$)

- When the clock goes high, the pre-charge PMOS turns off, and the bottom evaluation NMOS turns on. Now, the logic depends entirely on the inputs feeding the Pull-Down Network (PDN).
- The internal dynamic node can either stay HIGH (if the PDN is off) or discharge to LOW (if the PDN is on).
- **The Crucial Rule:** Because the final output started at 0 (from the pre-charge phase), it can only ever transition from **0 $\rightarrow$ 1** during evaluation. It is physically impossible for the final output to transition from **1 $\rightarrow$ 0** during this phase. This single-direction (monotonic) transition is exactly what prevents the glitching errors when these gates are cascaded!

Transistor Count Breakdown

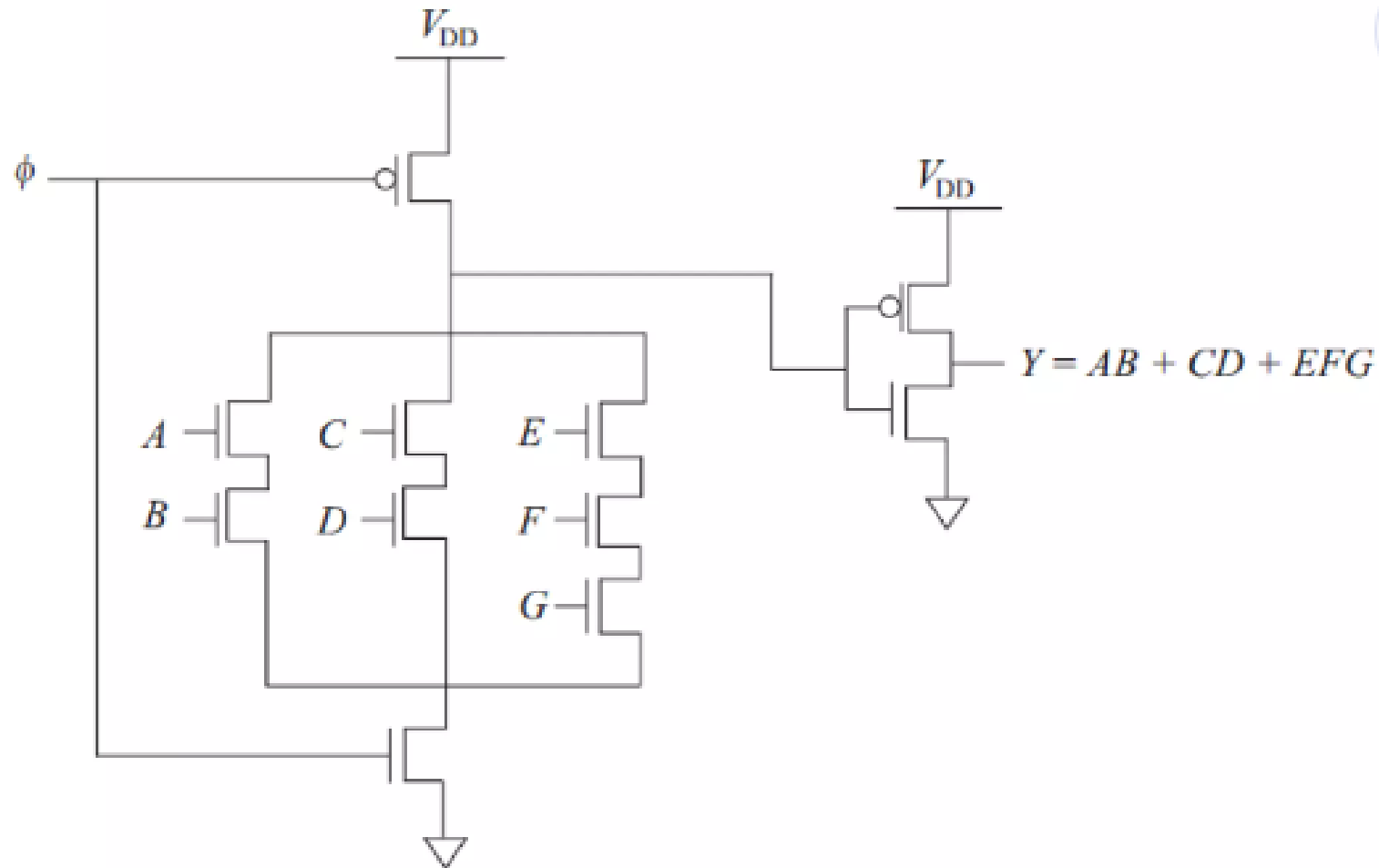
The slide also provides a great quick-reference for the hardware cost of an N -input logic function:

•**Static CMOS ($2N$):** Requires a full pull-up network (N PMOS) and a full pull-down network (N NMOS).

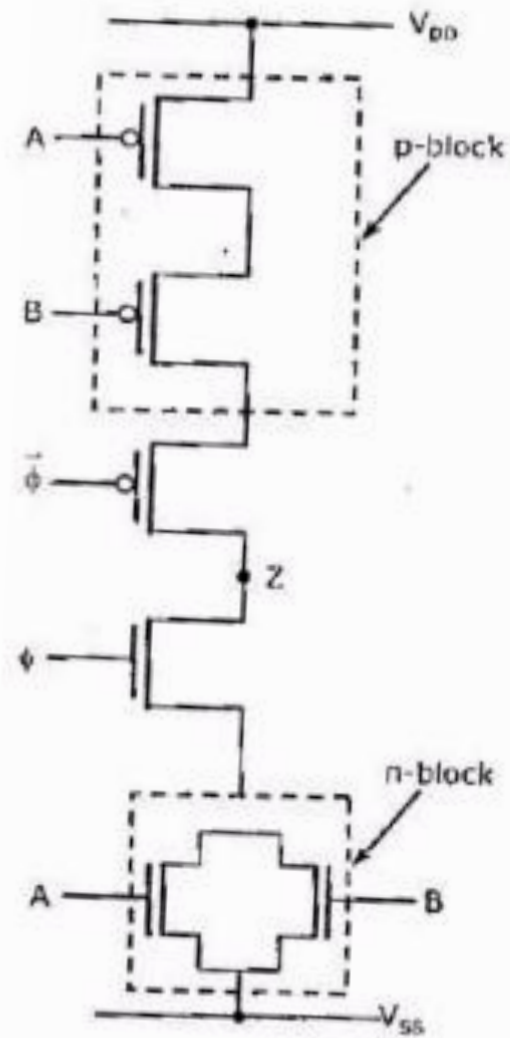
•**Dynamic CMOS ($N+2$):** Uses just the N NMOS transistors for the logic, plus 1 PMOS for pre-charge and 1 NMOS for evaluation. This saves a lot of space!

•**Domino CMOS ($N+2+2$):** Takes the Dynamic CMOS setup ($N+2$) and adds a standard static inverter (which consists of 1 PMOS + 1 NMOS = 2 transistors). So, it's slightly larger than pure dynamic logic, but still much smaller than static CMOS for complex gates.

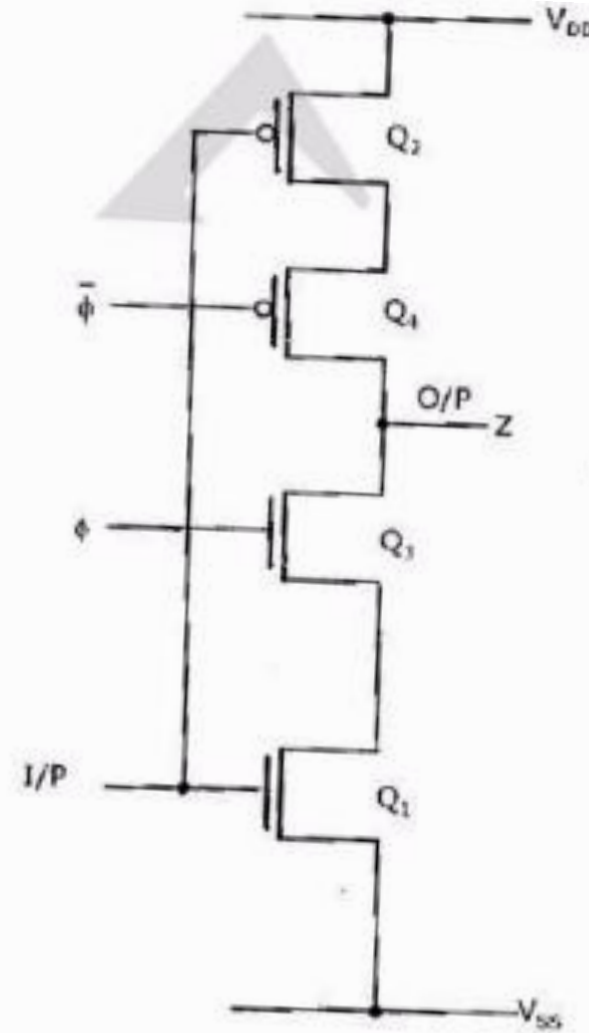
Example of CMOS Domino Logic(Dynamic + Inverter)



Clocked C² MOS Domino Logic (Dynamic + Inverter)



(a) Implemented CMOS Logic



(b) Clocked Inverter Circuit

Description

- A pull-up p-block and a complementary n-block pull-down structure represent p and n-transistors
- However, the logic in this case is connected to the output only during the ON period of the clock.

Working

- When $\phi = 1$ the circuit acts as an inverter, because transistors Q3 and Q4 are 'ON'.
 - It is said to be in the "Evaluation mode". Therefore the output Z changes its Previous value.
- When $\phi = 0$ the circuit is in hold mode, because transistors Q3 and Q4 are 'OFF'.
 - It is said to be in the "Pre Charge mode". Therefore the output Z remains its previous value.

Thank You!